

Requirements Traceability Matrix – Interview Integrated Enhancements

Requirement Summary Enhancements

ID	Original Requirement	Enhanced Interview-Based Addition
FR-01	Users can view a home page that explains the purpose of RoomRadar QC and links to room search.	Interview participants preferred using a dedicated website/application rather than manually searching CSV course files or external tools to find available rooms.
FR-02	Users can search for available rooms by day, start time, end time, duration, and building.	Student interviews validated this as the core workflow because many students explained difficulty finding available classrooms quickly between classes. Building/location filtering was especially important to reduce walking time across campus.
FR-03	The system validates that required search fields are provided.	Students emphasized the importance of fast and simple room searching, making validation necessary to prevent incomplete or confusing searches.
FR-04	The system validates that start time is before end time.	Interview feedback reinforced the need for accurate search results and proper room availability calculations.
FR-05	Users can view available room results with room code, building, and availability information.	Students wanted clear visibility into whether rooms were occupied, how long rooms would remain available, and where the rooms were located before entering them.
FR-06	Users can open a room detail page to view the room schedule for the selected day.	Interview participants wanted to know future classroom occupancy so they could avoid interruptions from upcoming classes.
FR-07	Search state is preserved during the current browser session.	Students explained that room searching often happens spontaneously during short breaks between classes, making quick navigation and preserved searches important.
FR-08	Admin users can log in with Firebase authentication.	Authentication protects administrative schedule upload functionality and prevents unauthorized users from modifying room data.
FR-09	Admin-only pages are protected from unauthenticated users.	Access control ensures only verified administrators can manage semester schedule uploads and room information.

FR-1 0	Admin users can log out and return to the public experience.	Session control improves security and allows administrators to safely end authenticated sessions.
FR-11	Admin users can choose semester season and year for uploaded data.	Semester organization supports accurate room availability calculations across different academic terms.
FR-1 2	Admin users can upload a CSV file containing course and room information.	Interview participants were unaware that course schedule CSV data existed, reinforcing the usefulness of automating schedule imports into the system.
FR-1 3	The backend stores semesters, rooms, and schedules in MySQL.	Persistent storage allows the application to maintain accurate room availability and scheduling data over time.
FR-1 4	The backend can search for rooms that do not conflict with scheduled classes.	Interviews confirmed that avoiding occupied classrooms was one of the most important user needs.
FR-1 5	The backend can return a room's schedule for a selected day.	Students wanted to know whether a room would remain available before deciding to study there.
FR-1 6	The backend handles database connection errors with a clear API response.	Reliability and troubleshooting support help maintain usability when backend services fail.
NFR- 01	The application provides a responsive web interface.	Most interview participants preferred using the system on mobile phones while moving around campus, reinforcing the need for responsive design.
NFR- 02	The system uses CORS to allow the frontend to call backend APIs during development.	Frontend/backend communication supports seamless development and testing workflows.
NFR- 03	Backend configuration can be controlled through environment variables.	Environment-based configuration improves deployment flexibility and protects sensitive credentials.

Traceability Matrix Enhancements

Req ID

Enhanced Interview-Based Traceability Addition

FR-01	Students preferred a dedicated platform over manually searching room schedules or CSV files.
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FR-02	Interviews validated building/location filtering because students wanted nearby rooms between classes.
FR-03	Validation supports the fast and convenient workflow students requested during interviews.
FR-04	Accurate time validation prevents invalid searches and incorrect room availability calculations.
FR-05	Students wanted room availability visibility before physically checking classrooms.
FR-06	Students requested schedule visibility to avoid entering rooms shortly before another class begins.
FR-07	Session persistence supports spontaneous room searching behavior between classes.
FR-08	Firebase authentication secures administrative room and semester management functionality.
FR-09	Protected routes prevent unauthorized access to administrative schedule uploads.
FR-10	Logout functionality supports proper session management and security practices.
FR-11	Semester controls help separate schedule data across academic terms.
FR-12	CSV uploads automate the process of importing classroom scheduling data.
FR-13	MySQL persistence supports long-term room and schedule management.
FR-14	Conflict-based searching directly addresses interview complaints about occupied rooms and wasted search time.
FR-15	Room schedule visibility helps students determine whether they have enough uninterrupted study time.
FR-16	Clear API/database error handling improves troubleshooting and reliability.
NFR-0 1	Responsive UI design supports mobile usage while students move around campus.
NFR-0 2	CORS configuration supports frontend/backend integration during development.
NFR-0 3	Environment variable configuration improves deployment flexibility and security.

Data Traceability Enhancements

Data Object

Enhanced Interview-Based Addition

semesters	Supports organization of schedule data across multiple semesters and academic terms.
rooms	Stores classroom information requested by students during interviews, including location and room identifiers.
schedules	Enables conflict checking and future availability calculations requested by students.
Firebase admin token	Protects administrative functionality from unauthorized access.
Search form state	Supports spontaneous and repeated room searching during short campus breaks.

Test Case Mapping Enhancements

Test ID	Enhanced Interview-Based Addition
TC-01	Verifies students can quickly access room searching from the homepage.
TC-02	Prevents incomplete searches that could confuse users.
TC-03	Ensures invalid time ranges do not return inaccurate room availability.
TC-04	Confirms room filtering supports efficient room discovery based on student interview needs.
TC-05	Confirms students can view future room occupancy before entering a classroom.
TC-06	Supports convenient navigation during spontaneous room searches.
TC-07	Ensures administrators can securely access upload functionality.
TC-08	Prevents unauthorized login attempts.
TC-09	Confirms protected administrative access control.
TC-10	Verifies proper logout and session cleanup functionality.
TC-11	Prevents invalid semester uploads.
TC-12	Confirms uploaded classroom schedules persist correctly in the database.
TC-13	Confirms backend reliability and troubleshooting support during database failures.

Known Gaps / Follow-Up Enhancements

Gap ID

Enhanced Interview-Based Impact

GAP-01	Search results may not reflect the real-time availability expectations students described during interviews.
GAP-02	Students may not receive the detailed schedule visibility requested during stakeholder interviews.
GAP-03	Missing backend admin protection could allow unauthorized schedule uploads.
GAP-04	Duplicate room insertion issues could affect schedule accuracy and room availability calculations.
GAP-05	Lack of automated tests increases the risk of regressions and undiscovered bugs.

Future Enhancement Ideas Mentioned During Interviews

- Live occupancy indicators
 - Reservation functionality
 - Notifications for room availability
 - Room images and classroom type information
 - Mobile-first UI improvements
 - Improved aesthetic design and interface polish
 - Live room availability updates
 - Student communication/collaboration features
 - Room usage counters/check-in systems
 - Future room planning and scheduling assistance
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Final Project Justification Statement

Student interviews were primarily used to validate and refine the usability of the original RoomRadar QC requirements rather than replace the existing system requirements. The interview feedback strongly aligned with the implemented features, particularly room searching, building filtering, schedule visibility, and mobile accessibility.